

HW 5 – Discrete and Continuous Random Variables

1) A random sample of adults was taken and asked the question “Last week, how many days did you sit down and eat at a restaurant?” X represents the number of days they responded with. Here is the resulting probability model:

Days (X)	0	1	2	3	4	5	6	7
Probability	0.40	0.28	0.16	0.05	0.04	0.03	0.02	0.02

(a) Show that the probabilities in this distribution add up to 1.

(b) Make a histogram of the probability distribution. Describe the shape of the distribution.

(c) Find and interpret $P(X < 4)$.

(d) Express the event “Ate out at least twice last week” in terms of X . What is the probability of this event?

(e) Find and interpret the mean.

(f) Find the standard deviation.

2) A Denver Ski Resort compiles their profit data yearly and has discovered that if it snows less than 60 inches in a season, they will lose \$90,000. If it snows between 60 inches and 80 inches, they will make \$70,000, and if it snows above 80 inches, they will make \$200,000. Let X represent the profit (any losses would be represented as a negative value). If the probability of less than 60 inches is 40%, the probability of between 60 inches and 80 inches is 45%, create a table to represent the probability distribution. Then comment on if this is a profitable business model or not based on the mean of the distribution.

3) A new type of gambling game uses two dice and the sum of the two dice determines your winnings. It costs \$5 to play the game. If the sum is a 2 or a 12, the player will win \$50. If the sum is a 7, the player will win \$10. Any other sum and the player will lose and not get anything. If X represents the winnings (with -5 representing a loss, since a player will lose that amount of money), create a table to represent the probability distribution. Using the mean of this distribution, do you believe the game is in favor of the player or the “house”?

4) Choose an American household at random and let the random variable X be the number of persons living in the household. The probability of X is:

X	1	2	3	4	5	6	7 or more
$P(X)$	0.25	0.32	0.17	0.15	0.07	0.03	0.01

Find and interpret the following probabilities:

a) $P(X > 3)$

b) $P(X \leq 4)$

c) $P(2 \leq x \leq 5)$